Two Entanglement Entropies of the Conjugate Weil Pair: the Admissible Pair Sector and the Full Residual Rank

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Abstract

The companion entanglement note and the orthogonality paper E1 identify two distinct entanglement entropies of the conjugate Weil pair $\{c, q-c\}$, which must not be conflated. The first is the entropy of the minimal admissible pair sector: the rank-3 admissible projection Π_q is spectrally atomic, equal to $\mathrm{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$; removing the zero mode leaves the two-dimensional Born–Infeld parity pair $\{e_{\xi_c}, e_{q-\xi_c}\}$, identified with the spin-$\frac{1}{2}$ representation. The proto-state on this sector is the singlet, and its reduced state is maximally mixed, so $S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$ for general canonical blocks, unconditionally relative to the cited upstream theorems. This is the sector relevant to the singlet correlator and the Tsirelson bound. The second is the full residual fibre-level rank $r_{\mathrm{pair}}(n) = R_\infty - R(n)$ defined on the full Gram–Schmidt basis (rank q at saturation, not the rank-3 admissible projection). E1 proves $S_{\mathrm{ent}}(n) = \log(r_{\mathrm{pair}}(n))$ unconditionally for matched single-character blocks; for independently sampled canonical blocks it remains conditional on the residual admissibility indiscernibility hypothesis $[\mathrm{H}_{\mathrm{res}}]$. The metaplectic dilation ϕ_ω is excluded as a bridge for $[\mathrm{H}_{\mathrm{res}}]$ because it acts between blocks, not within the residual support of one block.

Keywords: conjugate Weil pairs; spectral atomicity; admissible projection; spin-$\frac{1}{2}$ singlet; Tsirelson bound; residual rank; entanglement entropy; epistemic stratification.

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1 Introduction

The companion entanglement note [3] and the orthogonality paper E1 [5] establish results about the entanglement of a closed conjugate Weil pair $\{c, q-c\}$ in the spectral admissibility cascade. A careful reading shows that two genuinely different entanglement entropies are in play, attached to two different projections. Conflating them produces an apparent tension between an unconditional $\log 2$ and a depth-dependent $\log r_{\mathrm{pair}}(n)$.

This note separates them. The separation is not cosmetic: the two objects live on projections of different rank, have different physical roles, and have different epistemic status. One is the entropy of the minimal admissible sector, which is two-dimensional and carries the singlet relevant to Bell–Tsirelson correlations; it is unconditional for general blocks. The other is the entropy of the full residual Gram–Schmidt rank, a finer pipeline-level observable; it is unconditional only for matched blocks and otherwise rests on a bridge hypothesis.

2 Two distinct objects

The O-series pipeline stores two distinct objects per conjugate pair, and *O21* [7] (Remark 2.1) is explicit that they must not be identified.

- The full Gram–Schmidt basis is a unitary $q \times q$ matrix whose rows span all of $\mathbb{C}^q$ at saturation. Its cumulative admitted rank is $R(n)$, growing to $R_\infty = q$ at saturation, and the residual fibre-level rank is

  $$r_{\mathrm{pair}}(n) = R_\infty - R(n),$$

  which is non-increasing in n. This is the object of the companion note [3]. It is *not* the theoretical admissible projection.

- The theoretical admissible projection Π_q is the rank-3 projection onto the first three Gram–Schmidt directions $\{\phi_0, \phi_1, \phi_2\}$, identified through the H_{eff}-projection coordinates of *O21* [7], Remark 2.1. All spectral statements about Π_q concern this rank-3 object.

We therefore distinguish the admissible pair entropy $S_{\mathrm{ent}}^{\mathrm{adm}}$, defined on Π_q (Section 3), from the full residual-rank entropy $S_{\mathrm{ent}}(n)$, defined on the full Gram–Schmidt basis (Section 4).

3 The admissible pair sector: $S_{\text{ent}}^{\text{adm}} = \log 2$

The admissible sector is not an arbitrary residual subspace. Its structure is fixed by spectral atomicity.

Theorem 1 (Spectral atomicity, *O21* [7], Proposition 4.1)**.** *For each prime q and each conjugate pair $(c, q-c)$, the three admissible directions are pure Fourier modes,*

$$\phi_0 = e_0, \qquad \phi_1 = e_{\xi_c}, \qquad \phi_2 = e_{q-\xi_c},$$

with $e_\xi(k) = q^{-1/2} e^{2\pi i \xi k/q}$ and $\xi_c \in \{1, \ldots, \lfloor q/2 \rfloor\}$. Equivalently $\operatorname{ran}(\Pi_q^{(c)}) = \operatorname{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$.

This is an unconditional theorem of *O21*: it replaces the earlier coercivity and tightness hypotheses with proved statements, and is confirmed numerically to machine precision for $q \in \{29, 61, 101, 151\}$. Its proof rests on three closed steps: the k=3 Weil fingerprints are pure Fourier modes (the orthogonality result of E1 [5]); Gram–Schmidt preserves purity; and the admissible rank is exactly three [4], which with the Born–Infeld parity involution forcing e_{ξ_c} and $e_{q-\xi_c}$ to enter together fixes the triple.

The zero mode is removed by admissibility.

Lemma 2 (Mean-zero, *O21* [7], Lemma 3.1)**.** *For every $f \in \operatorname{ran}(\Pi_q^{(c)})$, the Fourier coefficient $\hat{f}(0) = 0$.*

Proof. The Weil representation ρ_c with $c \neq 0$ is irreducible with non-trivial central character, so the constant function (the trivial representation) is orthogonal to $\operatorname{ran}(\Pi_q^{(c)})$, giving $\langle \mathbf{1}, f \rangle = \hat{f}(0) = 0$. □

After removing e_0, the admissible sector is the two-dimensional Born–Infeld parity pair

$$\mathcal{A}_c := \operatorname{span}\{e_{\xi_c}, e_{q-\xi_c}\}, \qquad q - \xi_c \equiv -\xi_c \pmod q,$$

which we identify with the spin-$\frac{1}{2}$ representation $V_{1/2} \cong \mathbb{C}^2$. The two factors of the conjugate pair $\{c, q-c\}$ each carry a copy of $\mathcal{A}_c$, and the bipartite proto-state lives on $\mathcal{A}_c \otimes \mathcal{A}_c$.

Theorem 3 (Admissible pair entropy)**.** *The proto-state on $\mathcal{A}_c \otimes \mathcal{A}_c$ is the singlet*

$$|\Omega_c\rangle = \frac{1}{\sqrt{2}}\big(|\xi_c\rangle \otimes |q-\xi_c\rangle - |q-\xi_c\rangle \otimes |\xi_c\rangle\big).$$

Its reduced state is $\rho_c^{\mathrm{adm}} = \frac{1}{2}\mathbf{1}_2$*, and hence*

$$S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2.$$

Proof. By Theorem 1 and Lemma 2, the admissible sector of each factor is $\mathcal{A}_c \cong V_{1/2}$. By *Q3* [8], Theorem 4.1, Born–Infeld indiscernibility forces the proto-state to be invariant under the diagonal action on $V_{1/2} \otimes V_{1/2}$, and by Schur's lemma applied to the Clebsch–Gordan decomposition $V_{1/2} \otimes V_{1/2} = V_0 \oplus V_1$, the unique normalised invariant vector is the singlet $|\Omega_c\rangle$. The Schmidt decomposition of $|\Omega_c\rangle$ has two terms with equal coefficients $1/\sqrt{2}$, so $\rho_c^{\mathrm{adm}} = \frac{1}{2}\mathbf{1}_2$ and $S_{\mathrm{ent}}^{\mathrm{adm}} = -\mathrm{Tr}(\frac{1}{2}\mathbf{1}_2 \log \frac{1}{2}\mathbf{1}_2) = \log 2$. □

Theorem 3 uses only the admissible parity pair $\{e_{\xi_c}, e_{q-\xi_c}\}$, fixed by spectral atomicity and parity for any block parameters. It does not require a canonical identification of the full residual supports. The admissible pair entropy is therefore $\log 2$ for general canonical blocks, unconditionally relative to the cited upstream theorems (the present note imports the singlet identification from *Q3* and the atomicity from *O21* rather than re-deriving them internally).

4 The full residual rank: matched blocks and $[\mathrm{H}_{\mathrm{res}}]$

The full residual fibre-level rank $r_{\mathrm{pair}}(n) = R_\infty - R(n)$ is a different object, defined on the full Gram–Schmidt basis rather than on Π_q.

On matched single-character blocks, E1 [5] (Theorem 7) proves that the residual Schmidt spectrum is flat and that

$$S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n).$$

The proof uses the sign-reflected Fourier supports $S_{q-c}^{\mathrm{res}}(n) = -S_c^{\mathrm{res}}(n)$, which canonically identify the residual modes of the two factors and yield a maximally entangled residual state.

For independently sampled canonical blocks, E1 also shows that this literal mode-by-mode identification is not automatic. *O17* [1] supplies equality of the residual dimensions of the two factors, but not a canonical pairing. Universal admission orthogonality ($w_j = 1$, E1 Theorem 4) ensures each factor has an orthonormal residual basis, but orthonormality and equal dimension do not by themselves fix the joint state or force a flat marginal.

Definition 1 (Residual admissibility indiscernibility $[\mathrm{H}_{\mathrm{res}}]$)**.** *At BFS depth n, the residual fingerprint directions of a conjugate Weil factor are residually*

admissibility-indiscernible if no invariant carried by the admissible generators distinguishes them, equivalently if assigning unequal weights would constitute a premature admissible resolution of the open residual fibre.

Proposition 4 (Conditional flatness)**.** *Assume the closure of the conjugate pair as a pure object, and assume* $[\mathrm{H_{res}}]$*. Then the reduced state on the residual support is* $\rho_c^{\mathrm{res}}(n) = r_{\mathrm{pair}}(n)^{-1}\mathbf{1}_{r_{\mathrm{pair}}(n)}$*, and* $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ *for general canonical blocks.*

Proof. If unequal weights on the residual directions were admissibility-visible, they would define a decomposition of the open residual fibre that is stable under the admissible generators, i.e. a premature splitting, which is forbidden by Axiom A3 [2] and the non-factorisability lemma of the Foundation paper prior to projection locking. Under $[\mathrm{H_{res}}]$, no admissibility-visible invariant distinguishes the residual directions. Therefore the admissible reduced state cannot assign different weights to them without introducing extra structure not present in the admissibility data: the only admissibility-invariant reduced state is proportional to the identity. Together with closure (pure joint state) and the equal residual dimensions of *O17* [1], the reduced state is $\rho_c^{\mathrm{res}}(n) = r_{\mathrm{pair}}(n)^{-1}\mathbf{1}_{r_{\mathrm{pair}}(n)}$, giving $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$. □

Remark 1 (The dilation does not supply $[\mathrm{H_{res}}]$)**.** *The metaplectic dilation* $\phi_\omega(a, b, z) = (\omega a, \omega^{-1}b, z)$ *of* O31 *[6] forces* $\sigma_c(n) = \sigma_{\omega c}(n)$*, relating the block of character* c *to the block of character* ωc*. It acts between blocks, not within the residual support of a single block, and so does not permute the residual modes of one factor. It cannot make the residual modes admissibility-indiscernible, and is therefore not a bridge for* $[\mathrm{H_{res}}]$*.* $[\mathrm{H_{res}}]$ *has a status parallel to the conditional hypotheses [H-color] and [H-ext] elsewhere in the programme.*

5 Relation to the Bell–Tsirelson sector

The entanglement that Bell–CHSH experiments measure is carried by the spin-$\frac{1}{2}$ singlet. By Theorem 3, this is exactly the admissible pair sector, with $S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$. The singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ and the unconditional Tsirelson bound $|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}$ are derived from the same admissible structure in *Q1* [2] (Theorem 2.14 and Corollary 2.16). The admissible pair entropy is therefore the entanglement entropy of the state behind those results.

The full residual-rank entropy $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ is a finer, pipeline-level observable that tracks the contraction of the full Gram–Schmidt span

toward saturation. It is not the quantity measured by a Bell experiment, and it should not be identified with the singlet entropy. The two coincide only in the degenerate case $r_{\mathrm{pair}}(n) = 2$.

6 Status

E1 [5] proves universal admission orthogonality: each admitted k=3 Weil fingerprint is a Fourier mode, and a newly admitted frequency is orthogonal to the previous span. *O17* [1] proves equality of the conjugate pair ranks.

The admissible spin-$\frac{1}{2}$ pair sector is closed by spectral atomicity (*O21* [7], Proposition 4.1), the mean-zero lemma, and the singlet identification (*Q3* [8], Theorem 4.1). The zero mode is excluded and the two surviving non-zero frequencies form the parity pair $\{e_{\xi_c}, e_{q-\xi_c}\}$. Therefore

$$S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$$

for general canonical blocks, unconditionally within the established *Q3*/*O21*/*O29*/*O30* input chain (the rank-3 input of *O29* [4] itself carries the analytical backing of *O30* [6]).

By contrast, the full residual-rank statement

$$S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$$

is unconditional only for matched single-character blocks (E1 Theorem 7). For independently sampled canonical blocks it remains conditional on [$\mathrm{H_{res}}$], the residual admissibility indiscernibility hypothesis (Definition 1). The metaplectic dilation is excluded as a bridge for [$\mathrm{H_{res}}$] (Remark 1).

The separation is therefore:

$$\begin{aligned} \text{Bell–Tsirelson sector} &\implies S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2, \\ \text{full residual-rank sector} &\implies S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n). \end{aligned}$$

The first holds for general canonical blocks, unconditionally relative to the cited upstream theorems. The second holds unconditionally only for matched single-character blocks, and otherwise rests on [$\mathrm{H_{res}}$].

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